■ AmritChidiya (■■■■■■■■■■)

"Apni Sone Ki Chidiya Ko Phir Se Udaan Do" ■■

A full-stack, AI-powered multilingual voice and chat welfare assistant connecting Indian citizens with verified government schemes, scholarships, and cyber safety tools.

■ Complete Project Documentation & Team Division

Detailed project guides are available in the repository:

- ■ **PROJECT_DOCUMENTATION.md** — Full technical architecture guide, data flow, component hierarchy, security protocols, and installation guide.
- ■ **TEAM_ROLES_DIVISION.md** — Complete 4-member role ownership matrix with individual technical deliverables and USPs.

■ 4 Team Members & Individual USPs (Unique Selling Propositions)

Each team member spearheads a distinct technological pillar that powers AmritChidiya:

Team Member	Engineering Role & Focus	Member USP (Unique Selling Proposition)
Member 1	**Frontend UI/UX & Citizen Access Lead**	**Phygital Last-Mile Bridge & Citizen CSC Dossier Generator** Bridges the digital divide by generating physical, printable **CSC Scheme Dossiers** (`AC-CSC-XXXXXX`) with NPCI/DBT bank checklists & VLE operator sign-off boxes for village Jan Seva Kendras, coupled with a responsive Next.js 16 glassmorphism UI & live audio Talk Mode.
Member 2	**Backend API & Speech Processing Lead**	**Low-Latency Vernacular Audio Pipeline with Indian Accent DSP** Engineered an end-to-end voice pipeline using Groq Whisper Large v3 with phonetic domain prompting for Indian vernacular speech, background silence/hallucination suppression, and high-fidelity Microsoft Edge-TTS neural streaming in 5 Indian languages.
Member 3	**AI Agent, Scheme Logic & Cyber Safety Lead**	**Anti-Hallucination Welfare Match Engine & Domain Cyber Shield** Built the LangGraph dual-phase state machine ensuring 0% hallucinated welfare schemes, combined with the **Official Domain Shield & Scam Checker** that cryptographically verifies `.gov.in` / `.nic.in` URLs and shields citizens from WhatsApp fraud.
Member 4	**Database, Auth Security & DevOps Lead**	**Zero-Trust Hybrid Cloud Persistence with Bcrypt Salted Auth** Architected an enterprise-grade dual database layer (Supabase Cloud PostgreSQL + auto SQLite failover replica), hardened by salted **Bcrypt (12 rounds)**, signed **24-hour JWT tokens**, SlowAPI rate limiting, and production cross-origin deployment.

■ Core Feature Highlights

- ■ **Multilingual Conversational AI**: Native conversational support across **5 languages** (Hindi, Hinglish, English, Marathi, Tamil).
- ■■ **Hands-Free Talk Mode**: Full-duplex conversational voice mode powered by Groq Whisper Large v3 and Edge-TTS neural voices.
- ■ **Yojana Eligibility Calculator**: Interactive multi-criteria matrix dynamically evaluating State, Category, Income, and Profession to estimate annual financial benefits.
- ■■ **Cyber Shield & Fake Scheme Checker**: Instant link verification tool protecting citizens from fraudulent WhatsApp schemes by verifying `.gov.in` and `.nic.in` domains with 1930 Cyber Helpline integration.
- ■ **Printable CSC Dossier**: One-click printable PDF/dossier for village Common Service Centers (Jan Seva Kendra) with mandatory document checklists (Aadhaar, DBT/NPCI bank passbook).
- ■ **Direct Official Portal Links**: Verified links to official central and state portals (`scholarships.gov.in`, `scholarship.up.gov.in`, `pmkisan.gov.in`, `myscheme.gov.in`).
- ■ **Hardened Security & Privacy**: 12-round Bcrypt password hashing, HS256 JWT authorization, strict IDOR ownership checks, SlowAPI rate-limiting, and ephemeral audio cleanup.

■■ Enterprise Security & Privacy Matrix

AmritChidiya enforces defense-in-depth across 7 security and privacy layers:

Security Pillar	Code Implementation	Threat / Risk Mitigated
1. Credential Encryption	`bcrypt.gensalt(rounds=12)` in `database.py` with auto-migration from legacy SHA-256.	Rainbow table attacks, GPU brute-forcing, password database leaks.
2. Session & IDOR Defense	HS256 JWT tokens (24h expiry) + strict user-ownership validation on `/chats/{user_id}`.	Insecure Direct Object References, token spoofing, unauthorized chat access.
3. HTTP Security Headers	`SecurityHeadersMiddleware` enforcing `nosniff`, `DENY` clickjacking, XSS filter, strict referrer.	MIME sniffing, framing/clickjacking attacks, cross-origin referer data leaks.
4. DoS & Abuse Throttling	SlowAPI rate limits on `/signup` (5/min), `/login` (10/min), `/chat` (30/min), STT (20/min).	Bot registration floods, brute-force dictionary attacks, LLM API quota drain.
5. Voice Data Privacy	`tempfile.NamedTemporaryFile` + guaranteed `os.unlink()` cleanup in `finally` blocks.	Persistent citizen voice recordings on server disks, audio privacy leakage.
6. AI Anti-Hallucination	Canonical `SCHEME_URL_MAP` + `myscheme.gov.in` fallback; Whisper silence hallucination filter.	Fake welfare portals, phishing links, ghost queries generated on ambient room noise.
7. Citizen Anti-Fraud Shield	`ScamShieldModal` (4 Golden Rules + 1930 Cyber Helpline); guidance vs direct apply disclaimers.	WhatsApp advance fee scams, Aadhaar OTP theft, deceptive dark patterns.

■■ Technology Stack

- **Frontend**: Next.js 16 (App Router), React 19, TypeScript, Tailwind CSS, Lucide Icons, ReactMarkdown
- **Backend**: FastAPI (Python 3.14), LangGraph, LangChain, Groq LLM (`qwen/qwen3.8-27b`)
- **Speech & Audio**: Groq Whisper Large v3 (STT), Microsoft Edge-TTS (Neural Voice Synthesis)

- **Database & Auth**: Supabase Cloud (PostgreSQL) + Local SQLite3 Fallback, Bcrypt, Python-JOSE (JWT)
- **Security & Infrastructure**: SlowAPI (Rate Limiting), Starlette Security Headers, CORS Origin Regex

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■ Quick Start Guide

1. Backend Server

```
cd backend
python -m venv .venv
source .venv/bin/activate  # On Windows: .venv\Scripts\activate
pip install -r requirements.txt
python main.py
```

Backend runs on `http://127.0.0.1:8000`.

2. Frontend Application

```
cd frontend
npm install
npm run dev
```

Frontend runs on `http://localhost:3000`.